Overview of Big Data & Apache Hadoop

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Objectives

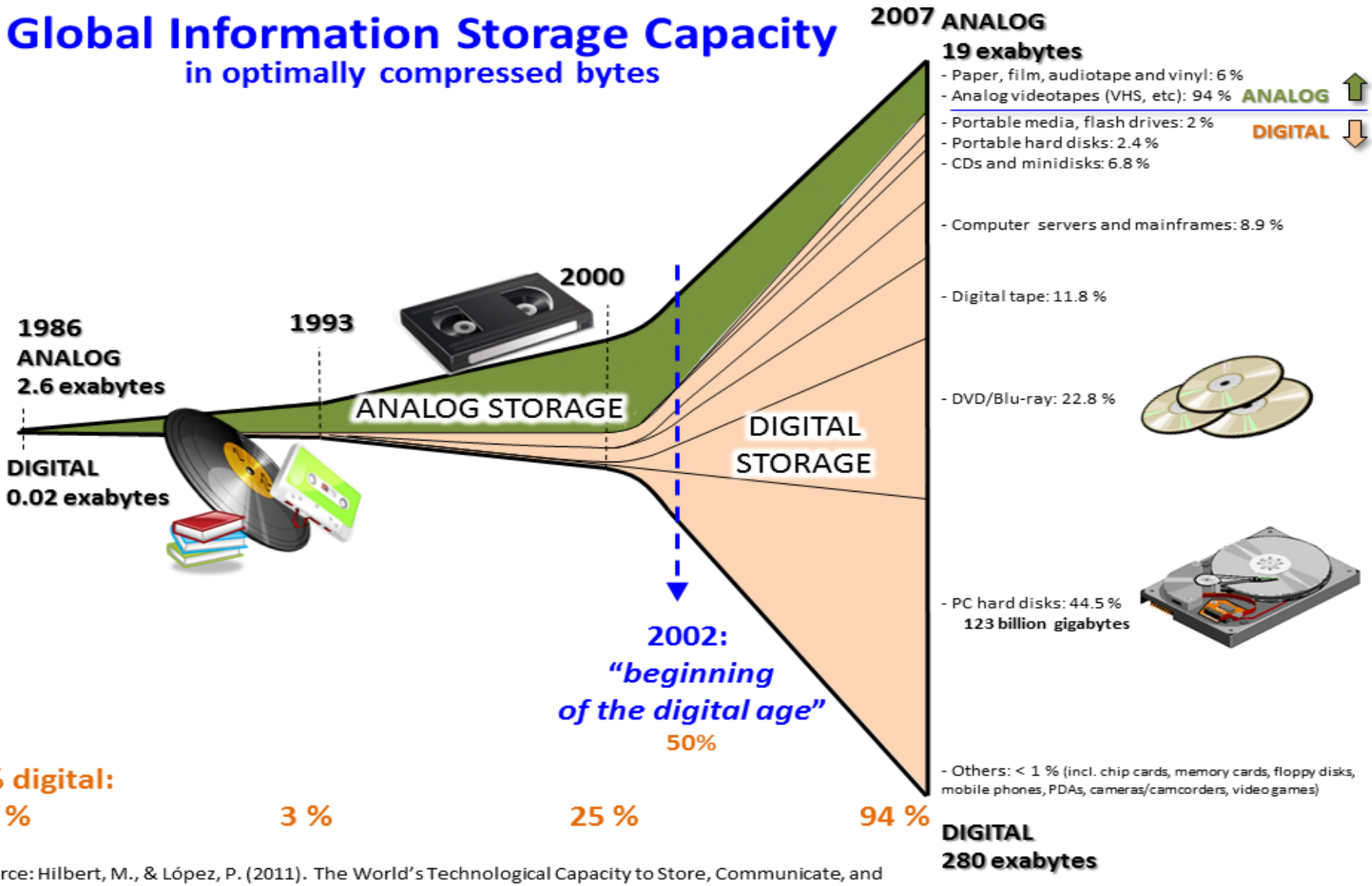
- What is Big Data
- BIG DATA Challenges
- Sources of Big Data challenges
- Categories Of 'Big Data'
- Characteristics Of 'Big Data’
- Live Example
- Introduction Of Hadoop

Big Data - Definition and Concepts

- **Big data** is the term for collection of data sets so large and complex that it become difficult to process using on – hand database management tool
- Traditionally, "Big Data" = massive volumes of data
  - E.g., volume of data at CERN, NASA, Google, …
- **Where does the Big Data come from?**
  - Everywhere! Web logs, GPS systems, sensor networks, social networks, Internet-based text documents, Internet search indexes, detail call records, astronomy, atmospheric science, biology, nuclear physics, biochemical experiments, medical records, scientific research, military surveillance, multimedia archives, …

Global Information Storage Capacity

in optimally compressed bytes



Source: Hilbert, M., & López, P. (2011). The World's Technological Capacity to Store, Communicate, and Compute Information. *Science*, 332(6025), 60 –65. http://www.martinhilbert.net/WorldInfoCapacity.html

Data Explosion

IBM 2.5 billion gigabytes of data was generated everyday in 2012.

Google 40,000 search queries search in every second.

YouTube 300 hours of video is uploaded every min.

facebook 31.25 million message sent & 2.77 million video been viewed.



2020 all data will pass through cloud .

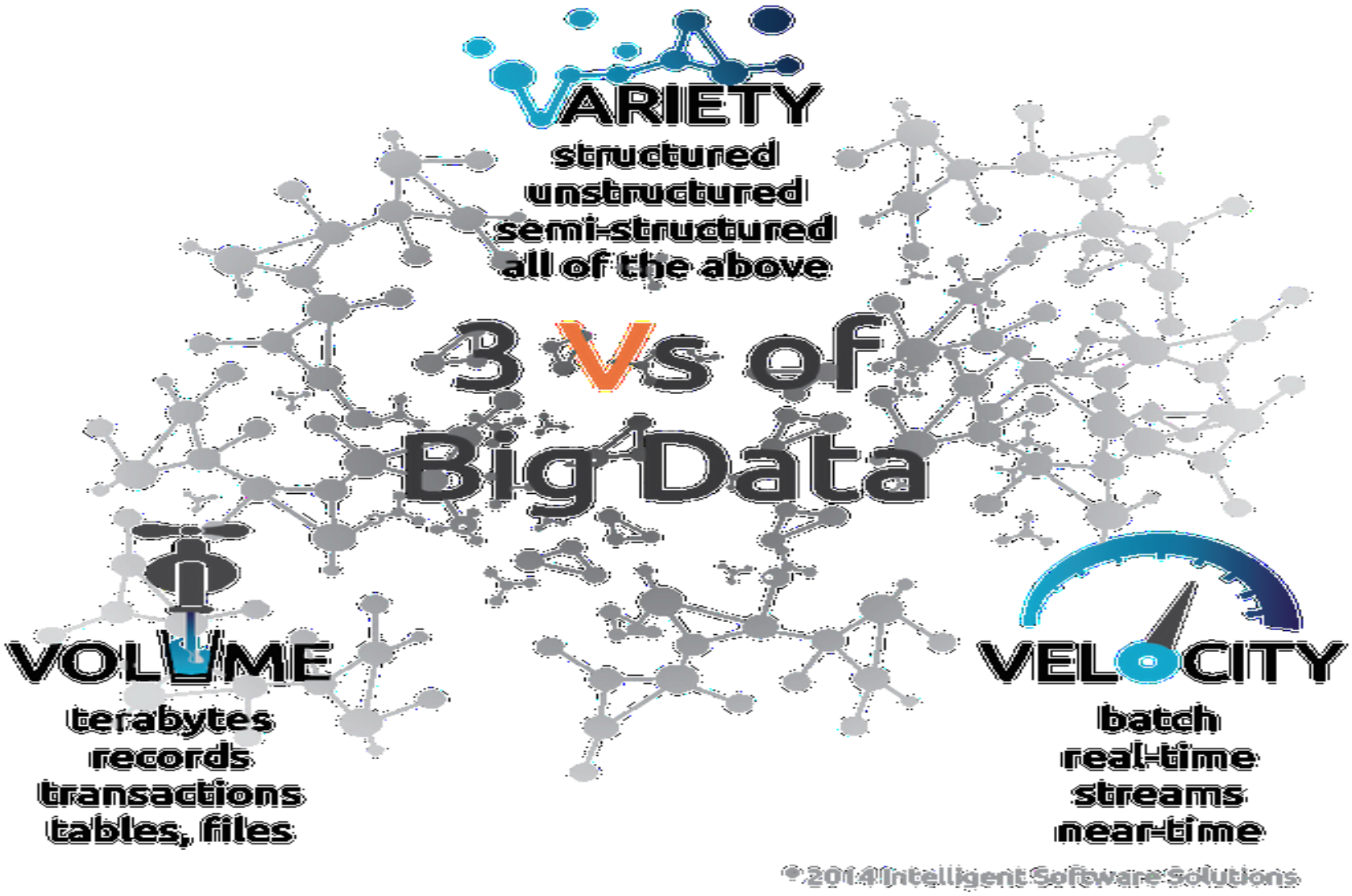
Technology Insights 6.1
The Data Size Is Getting Big, Bigger, …

- Hadron Collider - 1 PB/sec
- Boeing jet - 20 TB/hr
- Facebook - 500 TB/day
- YouTube – 1 TB/4 min
- The proposed Square Kilometer Array telescope (the world's proposed biggest telescope) – 1 EB/day

Name	Symbol	Value
Kilobyte	kB	10^3
Megabyte	MB	10^6
Gigabyte	GB	10^9
Terabyte	TB	10^{12}
Petabyte	PB	10^{15}
Exabyte	EB	10^{18}
Zettabyte	ZB	10^{21}
Yottabyte	YB	10^{24}
Brontobyte*	BB	10^{27}
Gegobyte*	GeB	10^{30}

*Not an official SI (International System of Units) name/symbol, yet.

Characteristics Of 'Big Data'



BIG DATA Challenges-

The challenges include :

- Capture,
- Storage,
- Search,
- Sharing,
- Transfer analysis and
- Visualization.

Categories Of 'Big Data'

Big data could be found in three forms:

1. **Structured**
2. **Unstructured**
3. **Semi-structured**

Structured

Any data that can be stored, accessed and processed in the form of fixed format is termed as a 'structured' data.

School Table

ID	Name
S001	University of Technology
S002	University of Applied Science

Student Table

School ID	ID	Name	DOB
S001	UT-1000	Tommy	05/06/1995
S001	UT-1000	Better	16/04/1995
S002	UAS-1000	Linda	02/09/1995
S002	UAS-1000	Jonathan	22/06/1995

Semi-structured Data

- In semi-structured data, the entities belonging to the same class may have different attributes even though they are grouped together.

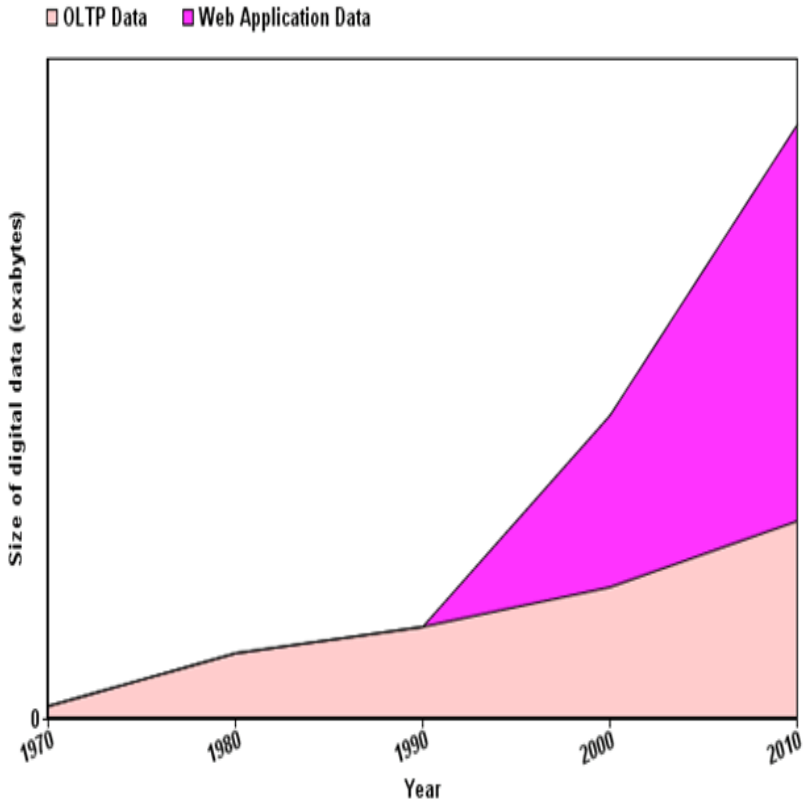
- Example :

  A Word document is generally considered to be **unstructured data**. However, you can add **metadata tags** in the form of **keywords** and other **metadata** that represent the document content and make it easier for that document to be found when people search for those terms -- the data is now **semi-structured**. Nevertheless, the document still lacks the complex organization of the database, so falls short of being fully **structured data.**

Semi-structured

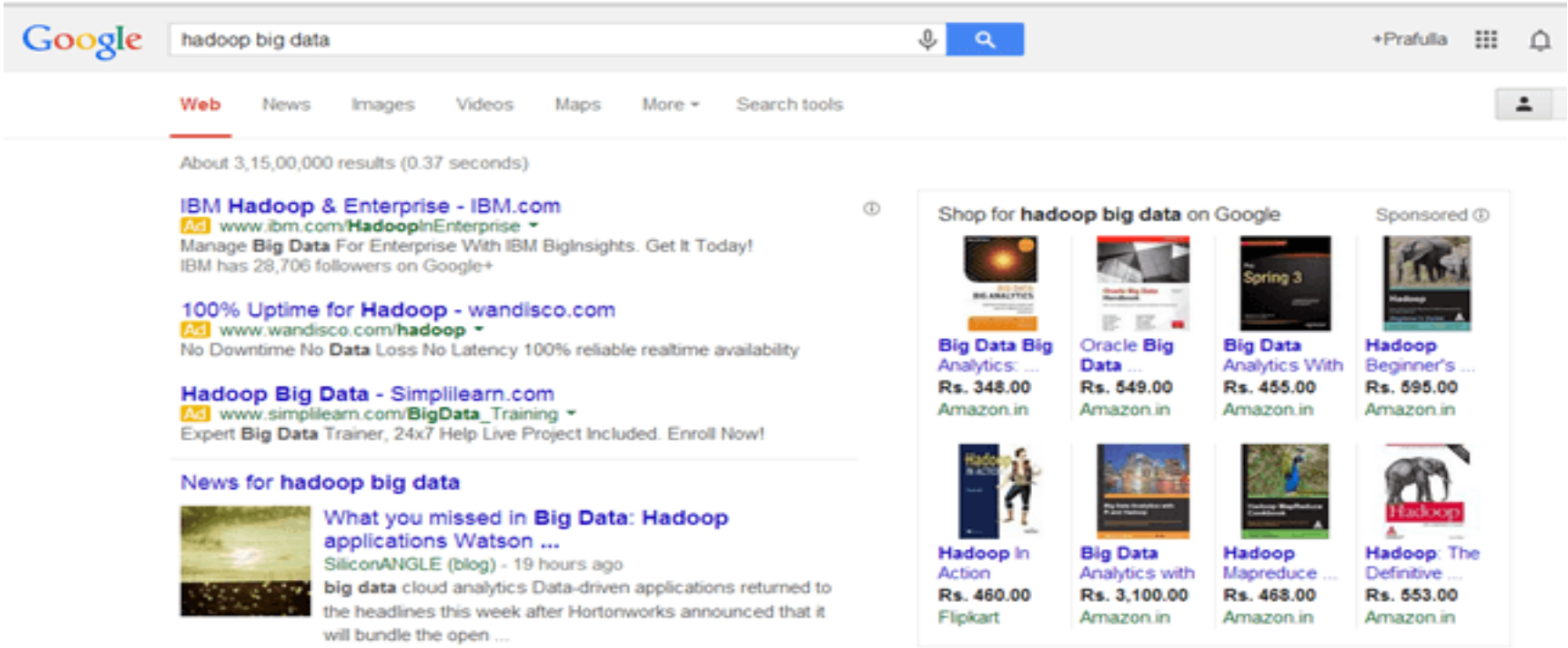
- **Semi-structured data can contain both the forms of data.**

```
<?xml version="1.0"?>
<catalog>
   <book id="bk101">
      <author>Gambardella, Matthew</author>
      <title>XML Developer's Guide</title>
      <genre>Computer</genre>
      <price>44.95</price>
   </book>
</catalog>
```



Unstructured

- Any data with unknown form or the structure is classified as unstructured data.
- Word, PDF, Text, Media Logs.



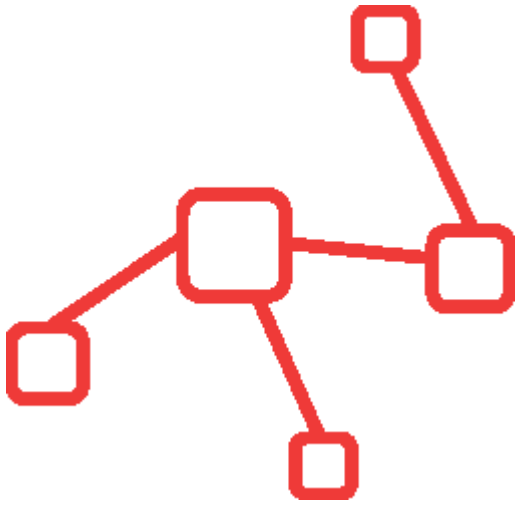
Live Example

Bank manager assigned task to find best location to setup the ATM machine .

Distributed System

- A model in which components located on networked computer communication.

- Distributed System use multiple Machine for a single job.



How does a distributed system works ?

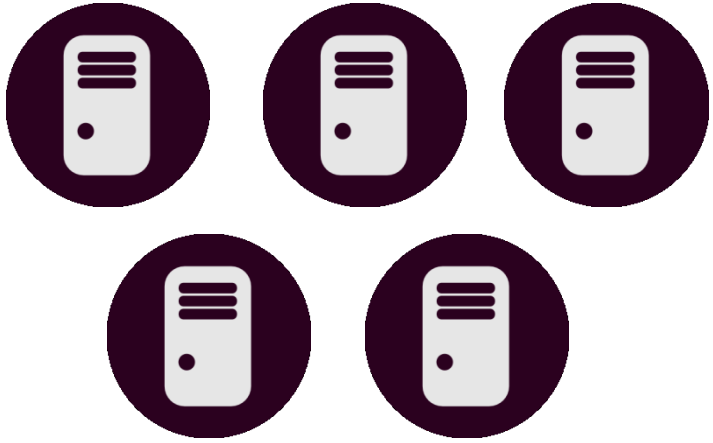
1 Machine



Data = 1 Terabyte

Processing time 45 min.

100 Machine

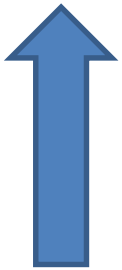


Data = 1 Terabyte

Processing time 47 sec.

Challenges of Distributed System

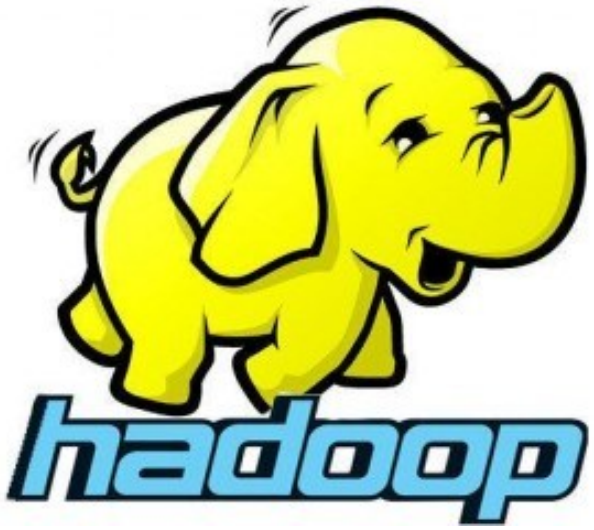
1. Multiple computer are used.
2. High chance of system failure.
3. Limit Bandwidth.
4. Complex Programming.



Solution to all these is Hadoop

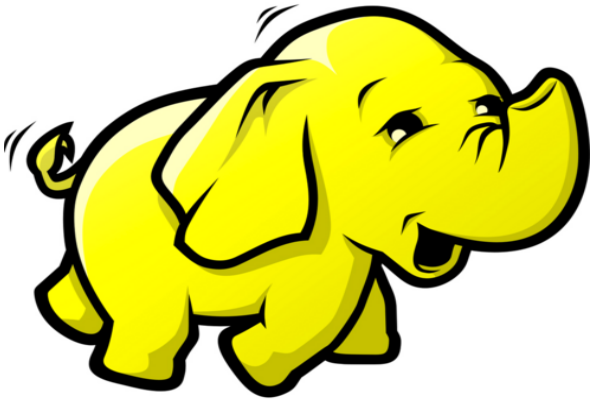
Big Data Technologies

- MapReduce …
- Yarn
- Hadoop …
- Hive
- Pig
- Hbase
- Flume
- Oozie
- Ambari
- Avro
- Mahout, Sqoop, Hcatalog, ….



Hadoop

- Doug Cutting is creator of Hadoop.
- Hadoop don’t have any meaning – its a made up name by his kid.





Hadoop

- Hadoop is a framework that allows for distributed processing of large data sets across clusters of commodity computers using simple programming models.



- Hadoop is an open source, java based programming framework that supports the processing and storage of extremely large data sets in a distributed computing environment.

Why Hadoop

1. **Runs a number of application which involving Petabyte of data.**
2. **Has a distributed file system, called HDFS, which enables fast data transfer among the nodes or server.**

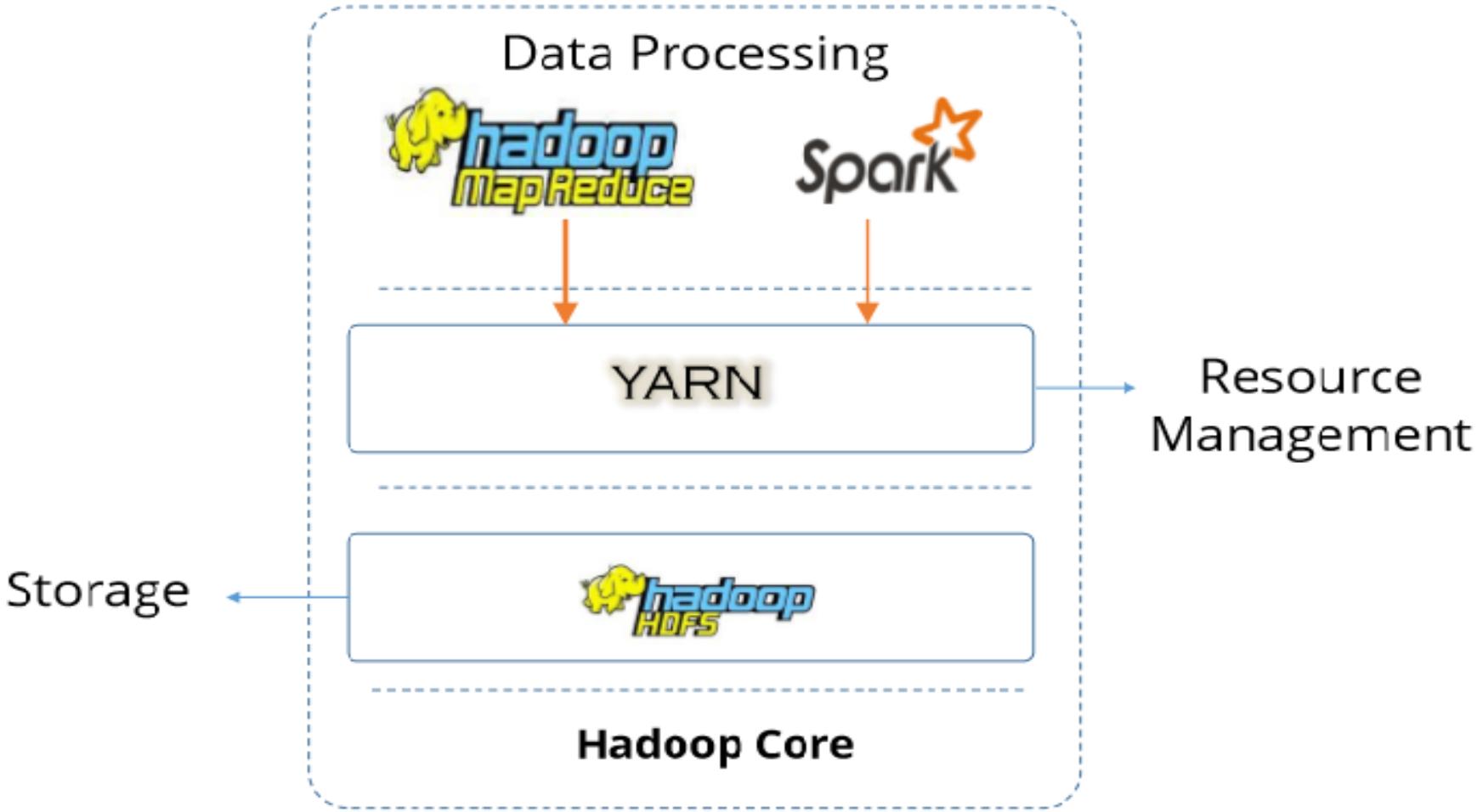
Big Data Technologies Hadoop

- Hadoop Technical Components
  - Hadoop Distributed File System (HDFS)
  - Name Node (primary facilitator)
  - Secondary Node (backup to Name Node)
  - Job Tracker
  - Slave Nodes (the grunts of any Hadoop cluster)
  - Additionally, Hadoop ecosystem is made up of a number of complementary sub-projects: NoSQL (Cassandra, Hbase), DW (Hive), …
    - NoSQL = not only SQL

Hadoop Characteristics

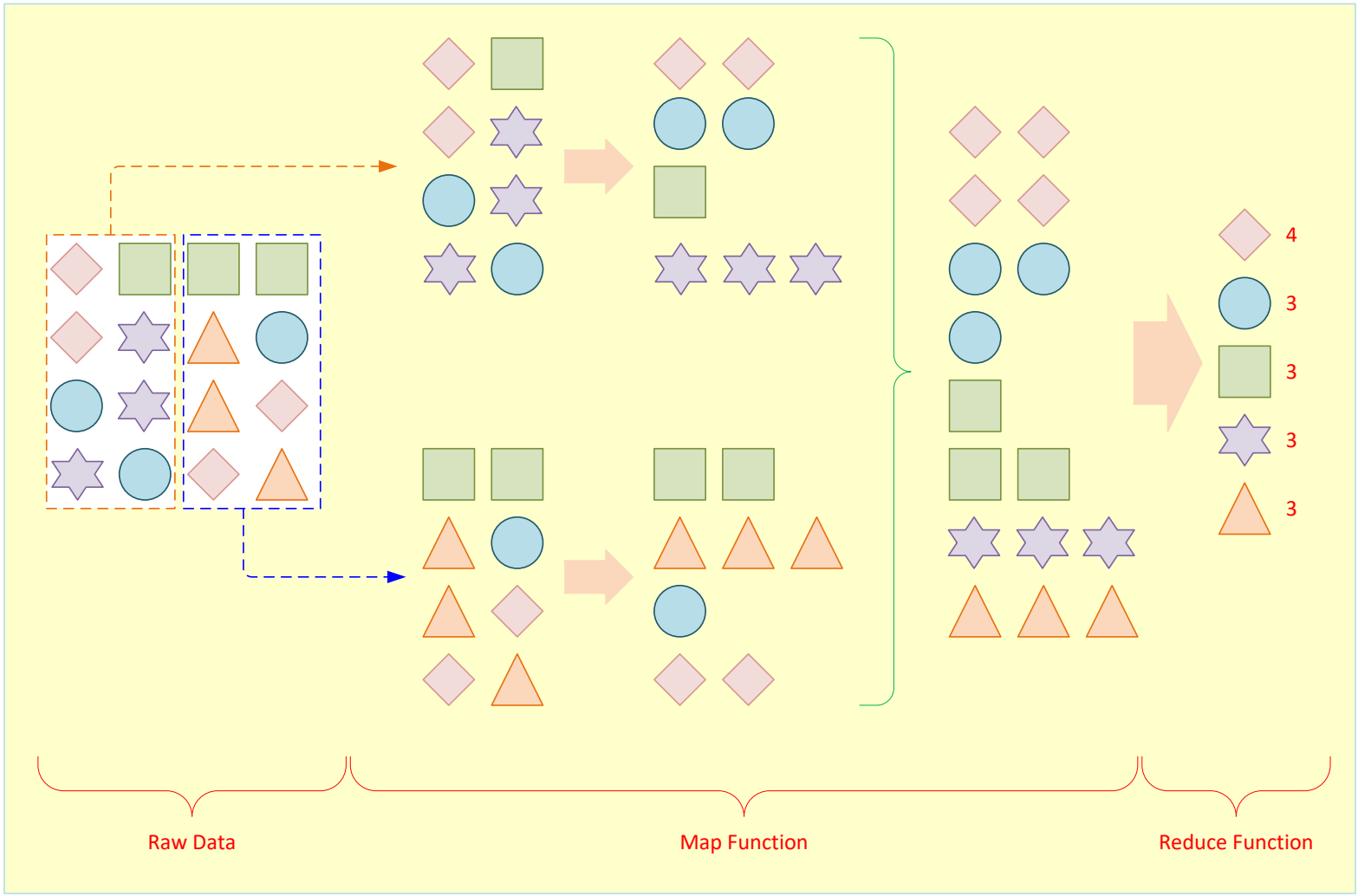
1. **Economical – ordinary computer can be used for data processing.**
2. **Reliable - Stores copies of the data on different machines and is resistant to hardware failure.**
3. **Scalable - Hadoop cluster can be extended by just adding nodes in the cluster.**
4. **Flexible - Can store a lot of data and decide to use it later.**

Hadoop Core Components



Big Data Technologies
MapReduce

How does MapReduce work?



Stream Analytics Applications

- e-Commerce
- Telecommunication
- Law Enforcement and Cyber Security
- Power Industry
- Financial Services
- Health Services
- Government

Thank You